Counterparty credit risk at a glance

Derivatives, SFTs and cleared trades in the EU under CRR3, with Basel references: SA-CCR, internal model method, CCPs, collateral and the latest developments

SA-CCR applies in the EU since **28 June 2021**. CRR3 keeps it and makes it the yardstick of the **output floor**. Supervisors now focus on CCR **management**: NBFI exposures, margining, stress testing and close-out.

KEY MILESTONES, 2005–2027

- **Jul 2005** — Basel II for trading: **IMM** and EPE (bcbs116)
- **Dec 2010** — **Basel III**: stressed EPE, wrong-way risk, CVA
- **Aug 2012** — **EMIR** in force; clearing and margin rules follow
- **Mar 2014** — **SA-CCR** replaces CEM (bcbs279)
- **Mar 2021** — **Archegos** default
- **Jun 2021** — SA-CCR applies in the EU (**CRR2**)
- **Dec 2024** — BCBS **CCR management** guidelines (d588)
- **Jan 2025** — **CRR3** applies; output floor
- **Jul 2025** — FSB **NBFI leverage** recommendations
- **Jan 2027** — EBA report on **SFT haircut floors** due

What is CCR and which method? CRR Art. 271–273b

The risk that the counterparty defaults **before final settlement** of the cash flows (Art. 272(1)). Exposure is **bilateral and uncertain**: it depends on future market values, netting and collateral.

Transactions	Methods (exposure value)
OTC and exchange-traded derivatives	SA-CCR (Art. 274–280f) · simplified SA-CCR (281) · OEM (282) · IMM (283–294)
SFTs, margin lending, long settlement	comprehensive method (223) · VaR (221) · IMM; long settlement: SA-CCR, simplified, OEM or IMM (273(8))
Cleared trades, default funds	CCP rules (Art. 300–311)

Exposure value × counterparty risk weight (SA or IRB) = RWEA; CVA risk is charged separately (Art. 381–386). Contractual netting recognised if legally enforceable (Art. 295–298).

Where exposure values are used CRR Art. 92, 384, 390, 429c, 429e

Use	Derivatives	SFTs
Own funds (CCR)	any method	any method
Output floor (S-TREA)	SA-CCR / simplified / OEM; no IMM	comprehensive method; no VaR, no IMM
CVA (BA-CVA)	CCR exposure value of the netting set (Art. 384(2)); SFTs only if fair-valued and material (Art. 382(2))	
Large exposures	SA-CCR / simplified / OEM; no IMM (Art. 390(4))	comprehensive method; IMM or VaR if the bank has permission for them (Art. 390(4), 401(2))
Leverage ratio	SA-CCR with multiplier 1 and only qualifying cash VM, except client trades cleared via a QCCP; simplified/OEM if used for own funds (Art. 429c)	gross assets + counterparty add-on $\max\{0; \sum E_i - \sum C_i\}$ (Art. 429e)

Latest developments status 30 October 2026 NEW

★ EU EUROPEAN UNION AND ITALY

- **1 Jan 2025**: CRR3 applies. SA-CCR essentially unchanged; the output floor uses SA-CCR, not the IMM, and SA-CCR there uses α = 1 instead of 1.4 for IMM derivatives until 31 Dec 2029 (Art. 92(5), 465(4)). EBA report on Basel minimum haircut floors for SFTs due **10 Jan 2027** (Art. 519d).
- **2024–26**: EMIR 3 (Regulation 2024/2987) in force 24 Dec 2024; active account requirement from 25 Jun 2025, detailed by RTS 2026/305 (in force 26 Feb 2026). Delegated Regulation 2025/855: SA-CCR delta for commodity options with negative prices.
- **ECB**: sound practices in CCR governance (Oct 2023); CCR remediation deadline end-2025 (priorities 2025–27); CCR exploratory scenario (Aug 2025: 15 banks; €96bn gross, €38bn net exposure to their selected top counterparties); CCR chapter of the internal models guide revised in v4.0 (Jul 2025; unchanged in v4.1, Jun 2026).
- **2026**: EBA central validation of ISDA SIMM (from 1 Mar 2026; SIMM reporting consultation to 2 Nov 2026); EBA Q&A 2025_7576: the Art. 207(2) correlation test applies to collateral under SA-CCR and IMM. Italy: TUF aligned to EMIR 3 (D.Lgs. 28/2026).

BASEL COMMITTEE AND FSB

- **11 Dec 2024**: BCBS **Guidelines for CCR management** (d588), replacing the 1999 sound practices: due diligence, margin, PFE and stress metrics, limits, close-out. **Jul 2025**: FSB NBFI leverage recommendations (Rec. 6: implement d588 timely and thoroughly); BCBS report on bank–NBFI interconnections (d598). **Feb 2026**: BCBS consults on consolidated guidelines; d588 becomes chapter CRI40.

UNITED KINGDOM

- Basel 3.1 (**PS1/26**) from 1 Jan 2027: SA-CCR **α = 1** for non-financial counterparties and pension schemes; revised SS12/13 on CCR.

UNITED STATES

- **19 Mar 2026** re-proposal: no advanced approaches (no IMM), wider SA-CCR netting, comprehensive-approach haircuts for repo-style trades; comments closed 18 Jun 2026, final rule pending.

SA-CCR · 1: exposure value CRR Art. 274–279c · CRE52 CRR2

$$\mathrm{EV} = \alpha\,(\mathrm{RC} + \mathrm{PFE}), \qquad \alpha = 1.4$$

Trade (δ × AdjNot × MF) → Hedging set (offsets, correlations) → Asset class (add-ons summed) → Netting set (multiplier × add-ons) → **EV** (1.4 × (RC + PFE))

Per **netting set**; α = 1.4 for all counterparties (Art. 274(2)). Margined EV ≤ unmargined EV (Art. 274(3)). EV = 0 for unmargined sets of sold options, premium received upfront, CMV always negative (Art. 274(5)).

1 Replacement cost (Art. 275)

$$\mathrm{RC}^{\mathrm{unmargined}} = \max\{\mathrm{CMV} - \mathrm{NICA};\ 0\}$$

$$\mathrm{RC}^{\mathrm{margined}} = \max\{\mathrm{CMV} - \mathrm{VM} - \mathrm{NICA};\ \mathrm{TH} + \mathrm{MTA} - \mathrm{NICA};\ 0\}$$

CMV: net market value before collateral; VM, NICA: net variation margin and net independent collateral, volatility-adjusted (+ received, − posted). Basel writes $\max\{V - C;\ TH + MTA - NICA;\ 0\}$ with C = VM + NICA.

2 Potential future exposure (Art. 278)

$$\mathrm{PFE} = \mathrm{multiplier} \times \textstyle\sum_a \mathrm{AddOn}^{(a)}$$

$$\mathrm{multiplier} = \min\left\{1;\ F + (1-F)\exp\left(\frac{z}{2(1-F)\sum_a \mathrm{AddOn}^{(a)}}\right)\right\}, \qquad F = 5\%$$

z = CMV − NICA, or CMV − VM − NICA if margined: excess collateral and out-of-the-money trades cut the PFE, to no less than 5% of the add-ons. a: IR, FX, credit, equity, commodity, other.

3 Trade level (Art. 279–279c)

$$\mathrm{RiskPosition}_i = \delta_i \times \mathrm{AdjNot}_i \times MF_i$$

ADJUSTED NOTIONAL (ART. 279B)

$$\text{IR, credit: } \mathrm{AdjNot} = \mathrm{notional} \times SD, \quad SD = \max\left\{\frac{e^{-0.05\,S} - e^{-0.05\,E}}{0.05};\ \frac{10}{\mathrm{OneBusinessYear}}\right\}$$

S, E: years to start and end date. FX: notional of the foreign leg (the larger, converted, if neither leg is in the reporting currency). Equity, commodity: units × price.

SUPERVISORY DELTA (ART. 279A)

$$\text{options: } \delta = \mathrm{sign} \cdot N\left(\mathrm{type} \cdot \frac{\ln\frac{P+\lambda}{K+\lambda} + 0.5\,\sigma^2 T}{\sigma\sqrt{T}}\right)$$

$$\text{linear: } \delta = \pm 1 \qquad \text{CDO tranches: } \delta = \pm\frac{15}{(1+14A)(1+14D)}$$

sign −1 for bought puts and sold calls; type +1 call, −1 put; T: latest exercise date; A, D: attachment and detachment points. ★ EU per-option λ: IR λ = max{0.10% − min{P;K}; 0}, σ = 50% (Delegated Regulation 2021/931); commodities λ = max{−1.1·min{P;K}; 0} (2025/855). Basel: one λ per currency.

MATURITY FACTOR (ART. 279C, 285)

$$MF^{\mathrm{unmargined}} = \sqrt{\frac{\min\{\max\{M;\ 10\ \mathrm{days}\};\ 1\ \mathrm{year}\}}{1\ \mathrm{year}}} \qquad MF^{\mathrm{margined}} = \frac{3}{2}\sqrt{\frac{\mathrm{MPOR}}{1\ \mathrm{year}}}$$

Margin period of risk (business days)	MPOR ≥
Daily-margined repos, securities lending, margin lending	5
★ EU Client–clearing member trades, either side (Art. 279c(1); Basel: member side)	5
Other margined netting sets	10
> 5,000 trades in the quarter, illiquid collateral, hard-to-replace derivatives	20
> 2 disputes longer than the MPOR in 2 quarters	× 2

Remargining every N days: MPOR = floor + N − 1 (Art. 285(5)). M, S, E in years, MPOR in business days; OneBusinessYear = one year in business days (Art. 279b–279c).

EU vs Basel at a glance CRR vs CRE51–56 ★ EU

Small books	Simplified SA-CCR and OEM (no Basel equivalent)
Hedging sets	5th commodity set for climatic conditions; 6th risk category "other" (SF 8%)
Negative rates	Per-option λ formula for IR and commodity options (Basel: bank-set λ per currency)
Output floor	SA-CCR α = 1 in S-TREA for IMM-modelled derivatives until end-2029 (Art. 465(4))
SFT floors	Basel minimum haircut floors not implemented (review 2027–28)
Client clearing	5-day MPOR also on the client side (Basel: clearing member only)

SA-CCR · 2: add-ons CRR Art. 277–280f · CRE52 CRR2

4 Hedging sets (Art. 277, 277a)

Each trade is mapped to the risk category of its **primary risk driver**, or to several categories if it has several material drivers (EBA RTS: Delegated Regulation 2021/931). Offsetting only within a hedging set.

Category	Hedging sets
Interest rate	one per currency; inflation trades in separate sets per currency
Foreign exchange	one per currency pair
Credit, equity	one each
Commodity	energy, metals, agricultural, other; ★ EU + climatic conditions
Other	one per identical risk driver

Basis trades and volatility/correlation trades form their own hedging sets; their SF is multiplied by ε = **0.5** and ε = **5** respectively (Art. 277a(2), 280).

5 Add-on by category (Art. 280a–280f)

INTEREST RATE: THREE MATURITY BUCKETS (< 1Y, 1–5Y, > 5Y)

$$\mathrm{AddOn}^{IR} = \textstyle\sum_j SF^{IR}\,\mathrm{EffNot}_j, \qquad \mathrm{EffNot}_j = \sqrt{D_1^2 + D_2^2 + D_3^2 + 1.4\,D_1D_2 + 1.4\,D_2D_3 + 0.6\,D_1D_3}$$

FOREIGN EXCHANGE

$$\mathrm{AddOn}^{FX} = \textstyle\sum_j SF^{FX}\left|\sum_{l\in j}\mathrm{RiskPosition}_l\right|$$

CREDIT, EQUITY: SYSTEMATIC + IDIOSYNCRATIC

$$\mathrm{AddOn} = \sqrt{\left(\textstyle\sum_k \rho_k\,\mathrm{AddOn}_k\right)^2 + \sum_k\left(1-\rho_k^2\right)\mathrm{AddOn}_k^2}, \qquad \mathrm{AddOn}_k = SF_k\,\mathrm{EffNot}_k$$

COMMODITY: PER HEDGING SET, OVER COMMODITY TYPES

$$\mathrm{AddOn}_j = \sqrt{\left(\rho\textstyle\sum_k \mathrm{AddOn}_k\right)^2 + \left(1-\rho^2\right)\sum_k \mathrm{AddOn}_k^2}, \quad \rho = 40\%$$

k: reference entity, index or commodity type; EffNot_k = Σ RiskPosition of the trades on k. Other risks: 8% × |EffNot| per hedging set.

6 Supervisory parameters (Art. 279a, 280a–280f)

Category	SF	ρ	σ (options)
Interest rate	0.5%	–	50%
Foreign exchange	4%	–	15%
Credit single name, CQS 1 / 2 / 3	0.38 / 0.42 / 0.54%	50%	100%
Credit single name, CQS 4 / 5 / 6	1.06 / 1.6 / 6.0%		
Credit index, IG / non-IG	0.38 / 1.06%	80%	80%
Equity single name / index	32 / 20%	50 / 80%	120 / 75%
Commodity electricity / other	40 / 18%	40%	150 / 70%
Other	8%	–	150%

★ EU credit factors by credit quality step (Basel: AAA–CCC, same values). Unrated names: IRB banks map internal ratings; SA banks use 0.54% (1.6% for high-risk items, Art. 128).

REMOVED CRR2 deleted the mark-to-market method (CEM) and the standardised method (SM); the original exposure method was revised. SA-CCR distinguishes margined from unmargined sets, recognises over-collateralisation and gives more risk-sensitive netting than the CEM.

Small derivative books CRR Art. 273a–273b, 281–282 ★ EU

Method	Derivatives ≤	Simplification
Simplified SA-CCR (Art. 281)	10% of assets and €300m	RC ignores collateral (max{CMV;0} or TH+MTA); multiplier 1; δ = ±1; SD = E − S; MF 1 (0.42 if margined); no offsets across buckets or names
Original exposure method (Art. 282)	5% of assets and €100m	EV = 1.4 × (RC + Σ notional × %); PFE × 0.42 if margined

OEM % of notional	IR	Credit	FX	Equity	Electricity	Gold, other commodities
Art. 282(4)	0.5% × M	6% × M	4%	32%	40%	18%

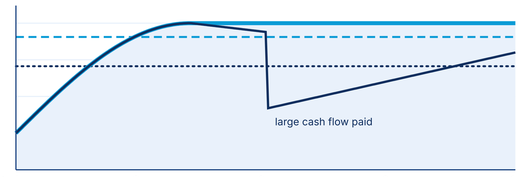
Size = |Σ long| + |Σ short| market values, tested monthly (Art. 273a(3)). Stop within 3 months after 3 consecutive months, or more than 6 of the last 12, in breach; re-entry after one clean year (Art. 273b). Internal-hedge credit derivatives excluded. OEM-eligible banks may set CVA own funds = CCR RWEA/12.5 (Art. 385). M: residual (or original) maturity in years.

Internal model method (IMM) CRR Art. 283–294 · CRE53 UNCHANGED

Permission for derivatives, SFTs and long settlement transactions; sequential roll-out allowed, return to standardised methods only with permission (Art. 283). Simulates the netting set's exposure distribution at future dates.

$$\mathrm{EV} = \alpha \times \text{Effective EPE}, \qquad \alpha = 1.4$$

1 Exposure profile (Art. 272, 284)



$$\mathrm{Eff.EE}_{t_k} = \max\{\mathrm{Eff.EE}_{t_{k-1}};\ \mathrm{EE}_{t_k}\} \qquad \text{Effective EPE} = \frac{\sum_k \mathrm{Eff.EE}_{t_k}\,\Delta t_k}{\min\{1\ \mathrm{year};\ \mathrm{maturity}\}}$$

EE: average exposure at a future date; $\mathrm{Eff.EE}_{t_0}$ = current exposure; sum up to one year or the longest maturity if shorter. IRB maturity M from the EE profile if the longest trade exceeds 1 year, capped at 5 years (Art. 162(2)(g)). Peak exposure (a high percentile of the distribution) is used for limits, not for own funds.

2 Calibration and α (Art. 284, 292)

- Own funds on the **higher** of Effective EPE with current data and with a **stress calibration** (3 years including a period of stressed counterparty CDS spreads); historical data ≥ 3 years, updated at least quarterly.
- Own estimate of α = economic capital from full simulation ÷ economic capital with EPE as a fixed EAD, floored at **1.2** (Art. 284(9)); supervisors may set α above 1.4 (Art. 284(4)).

3 Margined netting sets (Art. 285)

Model margining directly with permission; otherwise unmargined EEPE or the **shortcut**: potential increase in exposure over the MPOR + the larger of current exposure net of collateral and the largest exposure not triggering a call (threshold, MTA, initial margin). MPOR floors 5 / 10 / 20 days as in SA-CCR. Downgrade-triggered collateral is ignored.

4 Wrong-way risk (Art. 291)

- **General**: counterparty PD correlated with general market risk factors; captured through stress tests and scenarios.
- **Specific**: exposure correlated with the counterparty's own PD. With a **legal connection** between the counterparty and the underlying issuer: separate netting set; single-name CDS EAD = full expected loss on the underlying (LGD 100% under IRB); other single-name trades: jump-to-default (Art. 291(5); also under SA-CCR, Art. 273(9)).

5 Stress testing, use test, validation (Art. 286–294)

- **Stress tests**: principal risk factors at least monthly; multifactor scenarios at least quarterly (severe market events, liquidity drying up, liquidation by a large intermediary); joint stress of exposure and creditworthiness; reverse stress tests (Art. 290).
- **Use test**: model output integrated into credit approval, CCR management, capital allocation and governance, with at least one year of prior use (Art. 289). Independent CCR control unit, daily reporting, annual review (Art. 287, 293).
- **Backtesting** over several horizons out to at least one year, incl. typical MPORs; representative portfolios (Art. 294).

★ EU CRR3 keeps the IMM, but the output floor uses SA-CCR: TREA = max{U-TREA; 72.5% × S-TREA}, phased in from 50% (2025) and 55% (2026). In S-TREA, SA-CCR uses α = 1 until 31 Dec 2029 for derivatives modelled under the IMM (Art. 92(3), 92(5)(a), 465(1), 465(4)). ECB guide to internal models: CCR chapter revised in v4.0 (Jul 2025) on trade coverage, MPOR, collateral, IM modelling, maturity, stress calibration and α.

Central counterparties CRR Art. 300–311 · EMIR Art. 50a–50c · CRE54 UNCHANGED

Exposure	Qualifying CCP	Non-qualifying CCP
Trade exposures (Art. 306)	**2%**	corporate RW
Client via clearing member, protected (Art. 305(2))	2%	bilateral exposure to the clearing member (Art. 305(1))
Client not protected against joint default of member and another client (Art. 305(3))	4%	
Collateral posted, bankruptcy-remote	0 (Art. 306(2))	
Pre-funded default fund	K_i formula	1,250% (Art. 309)
Unfunded default fund	0% (Art. 310)	

$$K_i = \max\left\{K_{CCP}\,\frac{DF_i}{DF_{CCP} + DF_{CM}};\ 8\% \times 2\% \times DF_i\right\}, \qquad K_{CCP} = \textstyle\sum_i \mathrm{EAD}_i \times 20\% \times 8\%$$

RWEA = 12.5 × K_i (Art. 308). K_{CCP}: the CCP's hypothetical capital, computed at least quarterly with SA-CCR (MPOR 10 days) for derivatives and the comprehensive method for SFTs (EMIR Art. 50a–50b). DF_{CCP}: CCP's own prefunded resources; DF_{CM}: members' contributions. QCCP: authorised or recognised under EMIR (Art. 4(1)(88)). If a CCP stops qualifying, the bank has 3 months to move to the non-QCCP treatment (Art. 311). Clearing members' trade exposures and default fund contributions to QCCPs, and clients' trade exposures under Art. 305(2)–(3), are exempt from large exposure limits (Art. 400(1)(j), (l)).

SFTs, collateral and settlement CRR Art. 220–227, 271, 378–380 · CRE22, CRE56, CRE70 REVISED

Repos, securities or commodity lending and margin lending: comprehensive method, VaR approach (IRB only) or IMM (Art. 223, 221, 271(2)). Long settlement transactions (settlement later than the market standard or 5 business days after trade date, whichever is earlier): SA-CCR, simplified, OEM or IMM (Art. 272(2), 273(8)).

$$E^* = \max\{0;\ E(1 + H_E) - C\,(1 - H_C - H_{fx})\}$$

Comprehensive method (Art. 223), with maturity-mismatch adjustment. OWN ESTIMATES of haircuts deleted by CRR3 (Art. 225): supervisory haircuts only.

Supervisory haircuts, 10 days (Art. 224)	≤ 1y	1–3y	3–5y	5–10y	> 10y
Sovereign, CQS 1	0.5%	2%	2%	4%	4%
Sovereign, CQS 2–3	1%	3%	3%	6%	6%
Other issuers, CQS 1	1%	3%	4%	6%	12%
Other issuers, CQS 2–3	2%	4%	6%	12%	20%
Securitisation, CQS 1 / 2–3	2 / 4%	8 / 12%	8 / 12%	16 / 24%	16 / 24%

Sovereign CQS 4: 15%. Main-index equities 20%, other listed equities 30%, gold 20%, cash 0%, currency mismatch 8%. Liquidation period 10 days for margin lending and other capital-market transactions, 5 for repos and securities lending (÷ √2), 20 for secured lending (× √2); non-daily revaluation $H = H_M\sqrt{(N_R + T_M - 1)/T_M}$ (Art. 226).

$$\text{Netting sets: } E^* = \max\left\{0;\ \textstyle\sum_i E_i - \sum_j C_j + 0.4\,E_{net} + 0.6\,\frac{E_{gross}}{\sqrt{N}} + \sum_k \left|E_k^{fx}\right| H_k^{fx}\right\}$$

Master netting agreements (Art. 220, CRR3): $E_{net} = |\Sigma\ E_i H_i|$, $E_{gross} = \Sigma\ E_i|H_i|$, N = number of security issues and commodity types, excluding those below 1/10 of the largest. **VaR approach** (Art. 221): 99%, 5 or 10 days, ≥ 1 year of data; ★ EU CRR3 limits it to IRB exposures; excluded from the output floor. **0% haircut** for repos and securities lending of cash or 0%-RW sovereign debt with core market participants (Art. 227).

★ EU Basel **minimum haircut floors** for non-centrally cleared SFTs lending cash (or upgrading collateral) to unregulated counterparties against non-sovereign collateral (CRE56; below the floor: treated as unsecured) are **not in the CRR**: EBA report due 10 Jan 2027, Commission proposal by 10 Jan 2028 if appropriate (Art. 519d).

CRE56 floor	≤ 1y	1–5y	5–10y	> 10y	Main-index equities	Other
Corporate debt	0.5%	1.5%	3%	4%	6%	10%
Securitisations	1%	4%	6%	7%		

SETTLEMENT RISK (ART. 378–380)

Working days after due settlement (DvP)	5–15	16–30	31–45	≥ 46
% of price difference	8%	50%	75%	100%

Free deliveries: exposure from the first leg; 1,250% (or CET1 deduction) from 5 business days after the second leg was due (Art. 379); waiver for system-wide failures (Art. 380).

NEW new with CRR2 / CRR3 · REVISED changed · UNCHANGED earlier · REMOVED abolished · ★ EU differs from Basel

Abbreviations. CCP / QCCP (qualifying) central counterparty · CCR counterparty credit risk · CDS credit default swap · CEM current exposure method · CET1 common equity tier 1 · CM clearing member · CMV current market value · CQS credit quality step · CVA credit valuation adjustment · DF default fund · DvP delivery versus payment · EAD / EV exposure at default / exposure value · EE / EPE expected (positive) exposure · EMIR European Market Infrastructure Regulation · IG investment grade · IM / VM initial / variation margin · IMM internal model method · IR interest rate · IRB internal ratings-based approach · LGD loss given default · MPOR margin period of risk · MTA minimum transfer amount · NBFI non-bank financial intermediary · NICA net independent collateral amount · OEM original exposure method · PD probability of default · PFE potential future exposure · RC replacement cost · RW(EA) risk weight (risk-weighted exposure amount) · SA standardised approach · SA-CCR standardised approach for CCR · SD supervisory duration · SF supervisory factor · SFT securities financing transaction · SIMM ISDA Standard Initial Margin Model · FX foreign exchange · MF maturity factor · OTC over-the-counter · RTS regulatory technical standards · S-/U-TREA standardised / unfloored total risk exposure amount · TH threshold · VaR value at risk · WWR wrong-way risk.

Sources: Regulation (EU) No 575/2013 (CRR) as amended by Regulations (EU) 2019/876 and 2024/1623; Regulation (EU) No 648/2012 (EMIR) and Regulation (EU) 2024/2987; Delegated Regulations (EU) 2021/931, 2025/855 and 2026/305; D.Lgs. 28/2026; EBA Q&A 2025_7576; ECB sound practices in CCR governance (2023), CCR exploratory scenario (2025), guide to internal models (2025–26); BCBS CRE22, CRE51–56, CRE70, bcbs116, bcbs189, bcbs279, d588, d598; FSB NBFI leverage report (2025); PRA PS1/26 and SS12/13; US agencies' proposal of 19 March 2026. Simplified overview; status October 2026.